Industrial Internship Report on

"Student Management System"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

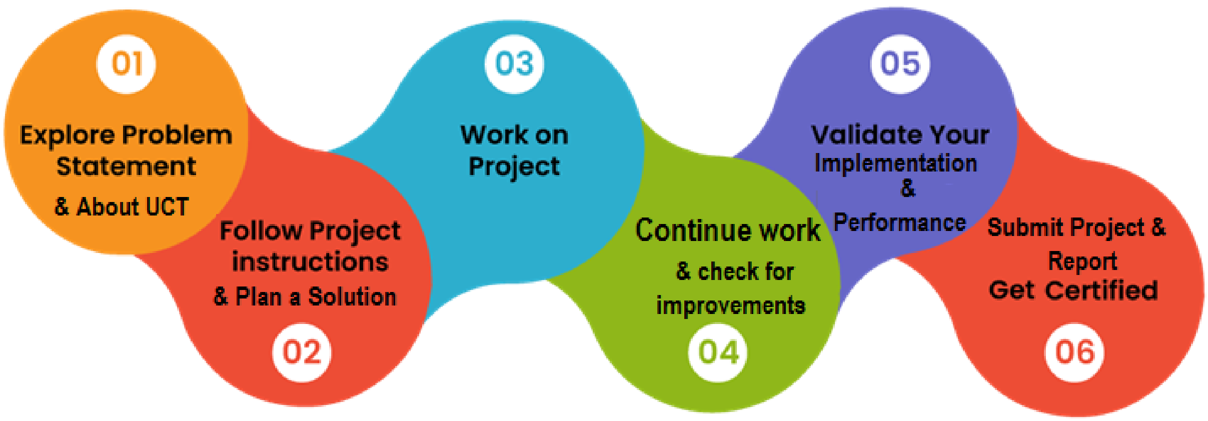
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

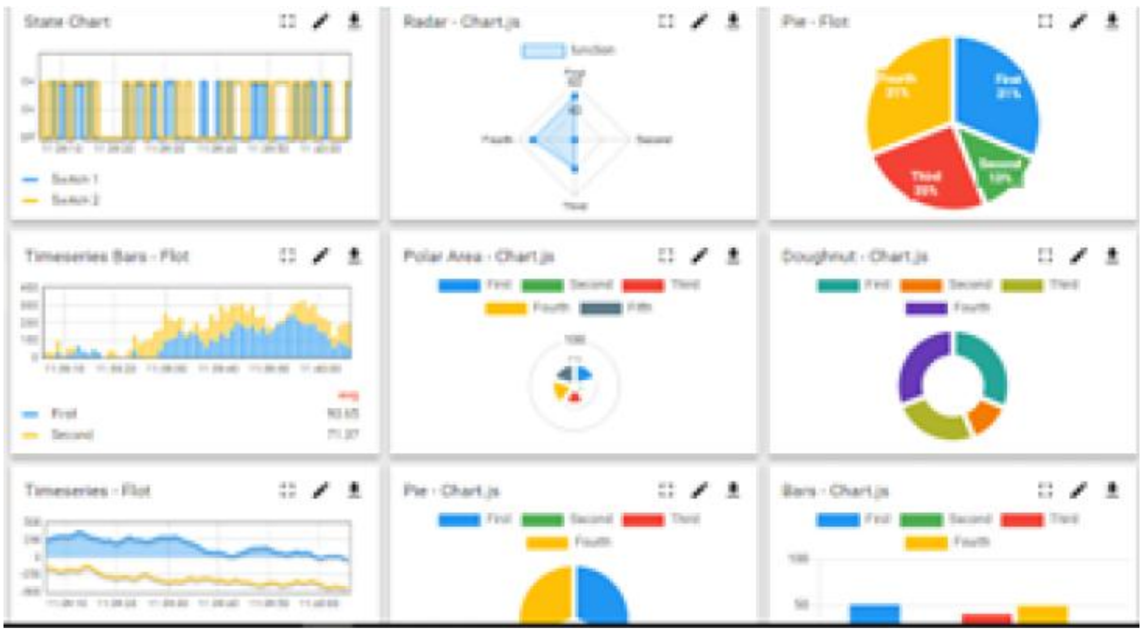
i. UCT IoT Platform (uct Insight)

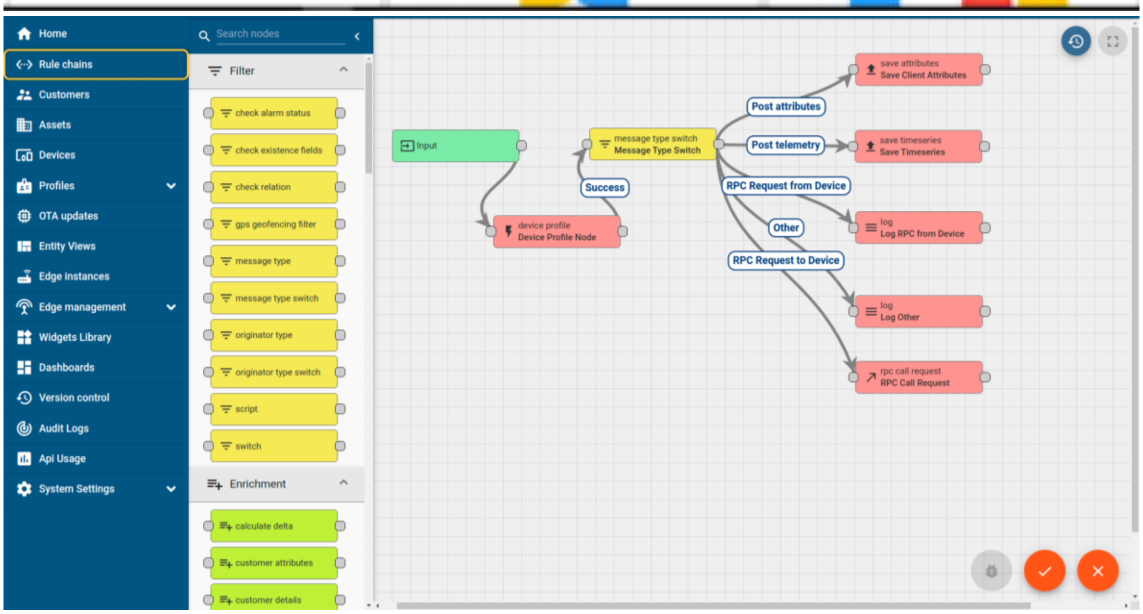
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

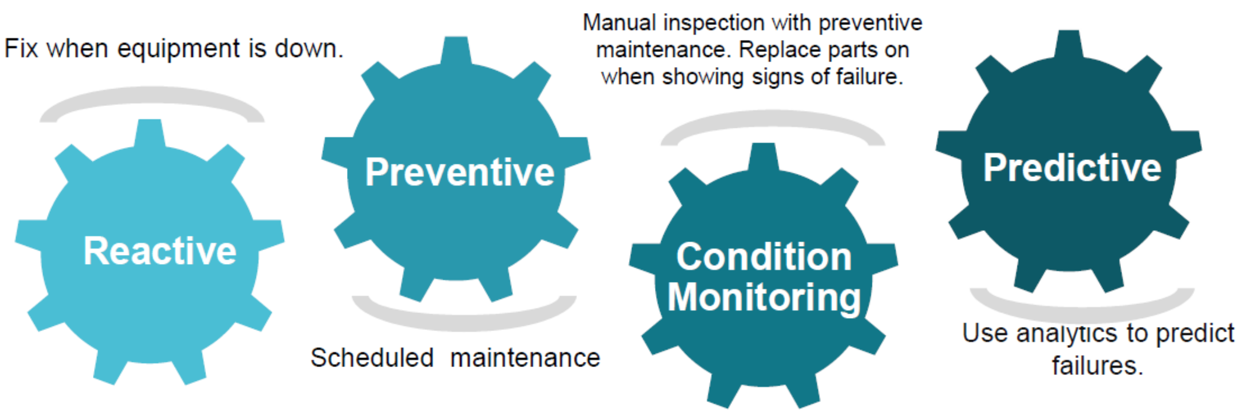
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

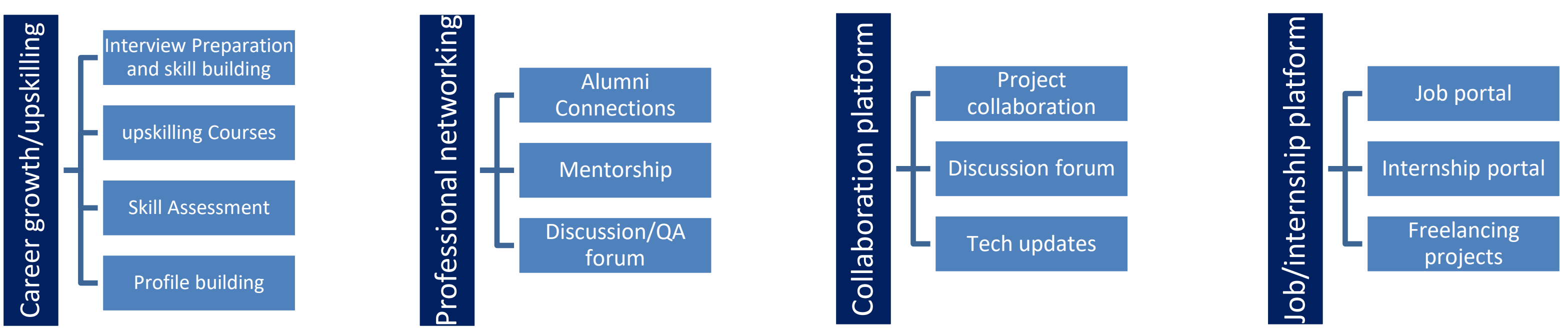
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] [1] Python Software Foundation, "Python Documentation"

https://docs.python.org/3/

[2] Flask Documentation

https://flask.palletsprojects.com/

[3] Mozilla Developer Network (MDN) Web Docs - HTML, CSS & JavaScript

https://developer.mozilla.org/

[4] W3Schools Web Development Tutorials

https://www.w3schools.com/

2.6 Glossary

Terms	Acronym
Student Management System	SMS
Object-Oriented Programming	OOP
Command Line Interface	CLI
Application Programming Interface	API
JavaScript Object Notation	JSON
HyperText Markup Language	HTML
Cascading Style Sheets	CSS
Flask Web Framework	Flask

3 Problem Statement

Educational institutions often face challenges in maintaining student records efficiently due to manual record keeping, spreadsheet dependency, and fragmented data storage methods. These approaches are prone to human errors, duplication, data inconsistency, and difficulty in retrieving information quickly.

The objective of this project is to design and develop a Student Management System (SMS) that provides a centralized platform for storing, managing, and retrieving student information. The system allows administrators to perform operations such as adding, updating, deleting, searching, and sorting student records while ensuring data integrity through validation mechanisms.

The proposed solution provides both a Command Line Interface (CLI) and a modern Web Dashboard, enabling users to manage student records through multiple interfaces while maintaining a common backend and database. The system reduces administrative effort, improves data accessibility, and demonstrates the practical application of Object-Oriented Programming, Flask web development, file handling, and software testing concepts.

4 Existing and Proposed solution

Existing Solutions:

Many small educational institutions still manage student records using paper-based registers or spreadsheet software such as Microsoft Excel. While these methods are simple, they suffer from several limitations:

- Lack of centralized data management.
- High chances of duplicate records.
- Difficult searching and filtering.
- No automated validation mechanisms.
- Limited scalability.
- Increased risk of accidental data loss.

Proposed Solution:

The proposed Student Management System addresses these limitations through a centralized software solution developed using Python and Flask.

Key features include:

- Add, update, search, and delete student records.
- Automatic validation of email, marks, age, and unique IDs.
- REST API based backend architecture.
- Interactive web dashboard.
- Command-line interface support.
- Statistical analysis of student records.
- Automated testing and verification.

Value Addition:

- Dual interface support (CLI + Web).
- Real-time data synchronization.
- User-friendly dashboard.
- Extensible architecture for future enhancements.
- Automated validation and error handling.
- Modular OOP design for maintainability.

4.1 Code submission (Github link)

https://github.com/profound307/upskillcampus

4.2 Report submission (Github link) :

5 Proposed Design/ Model

5.1 High Level Diagram (if applicable)

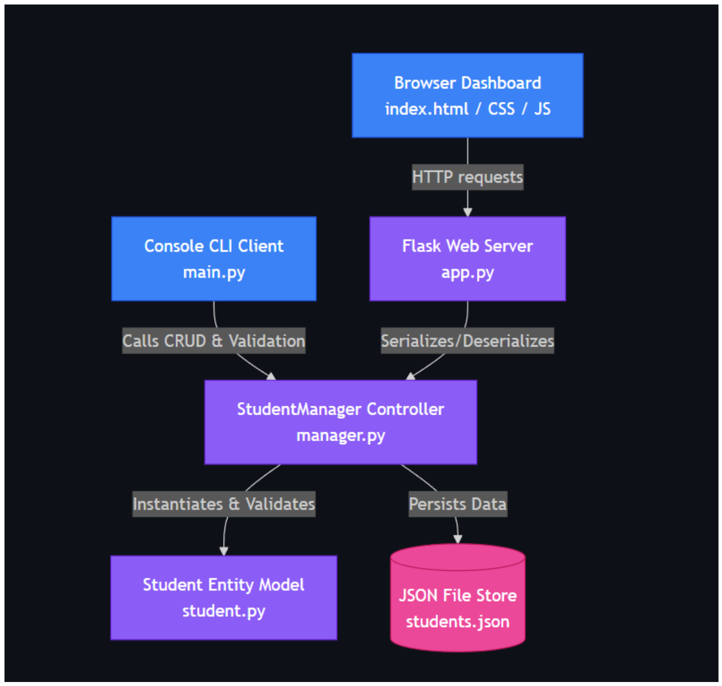
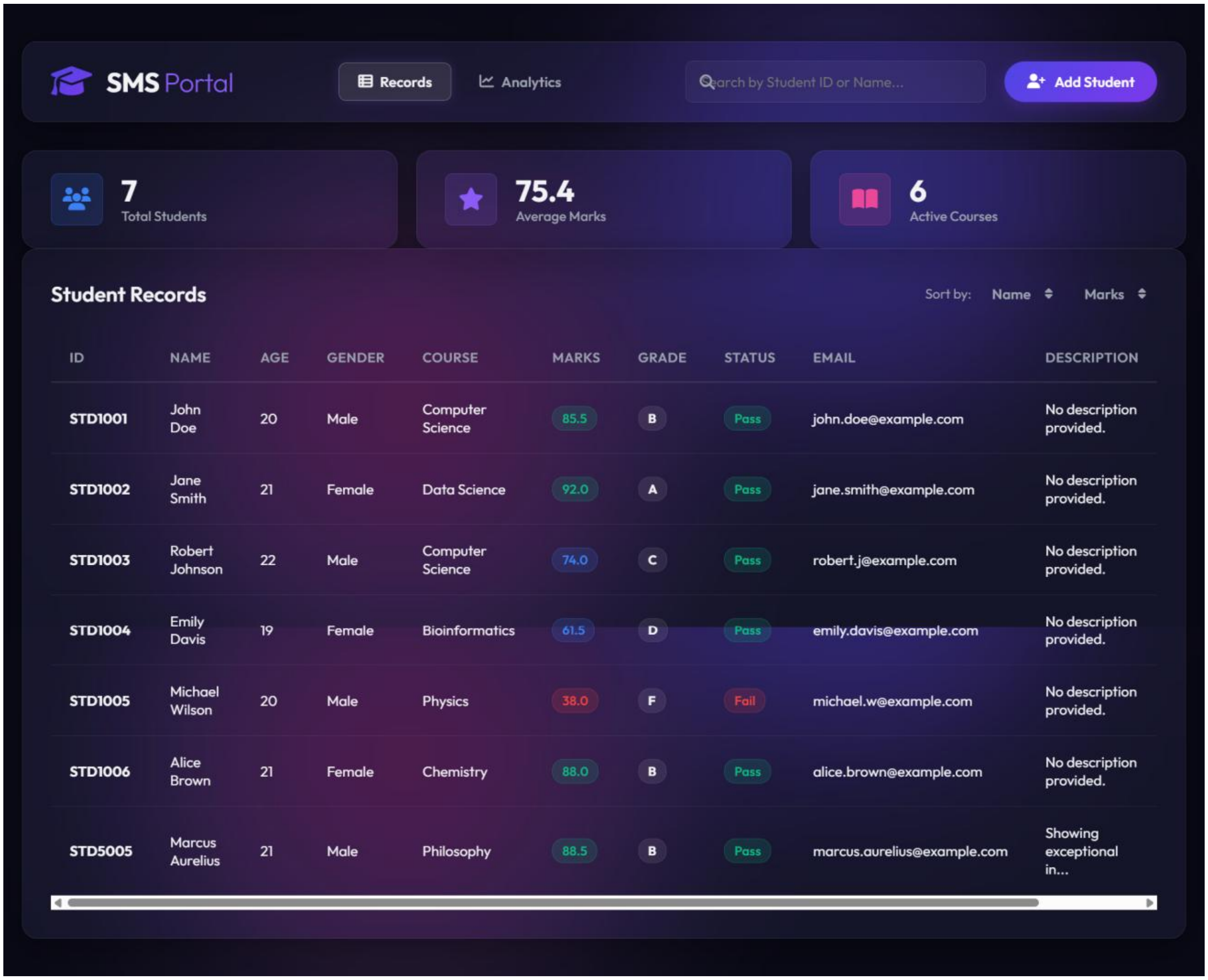
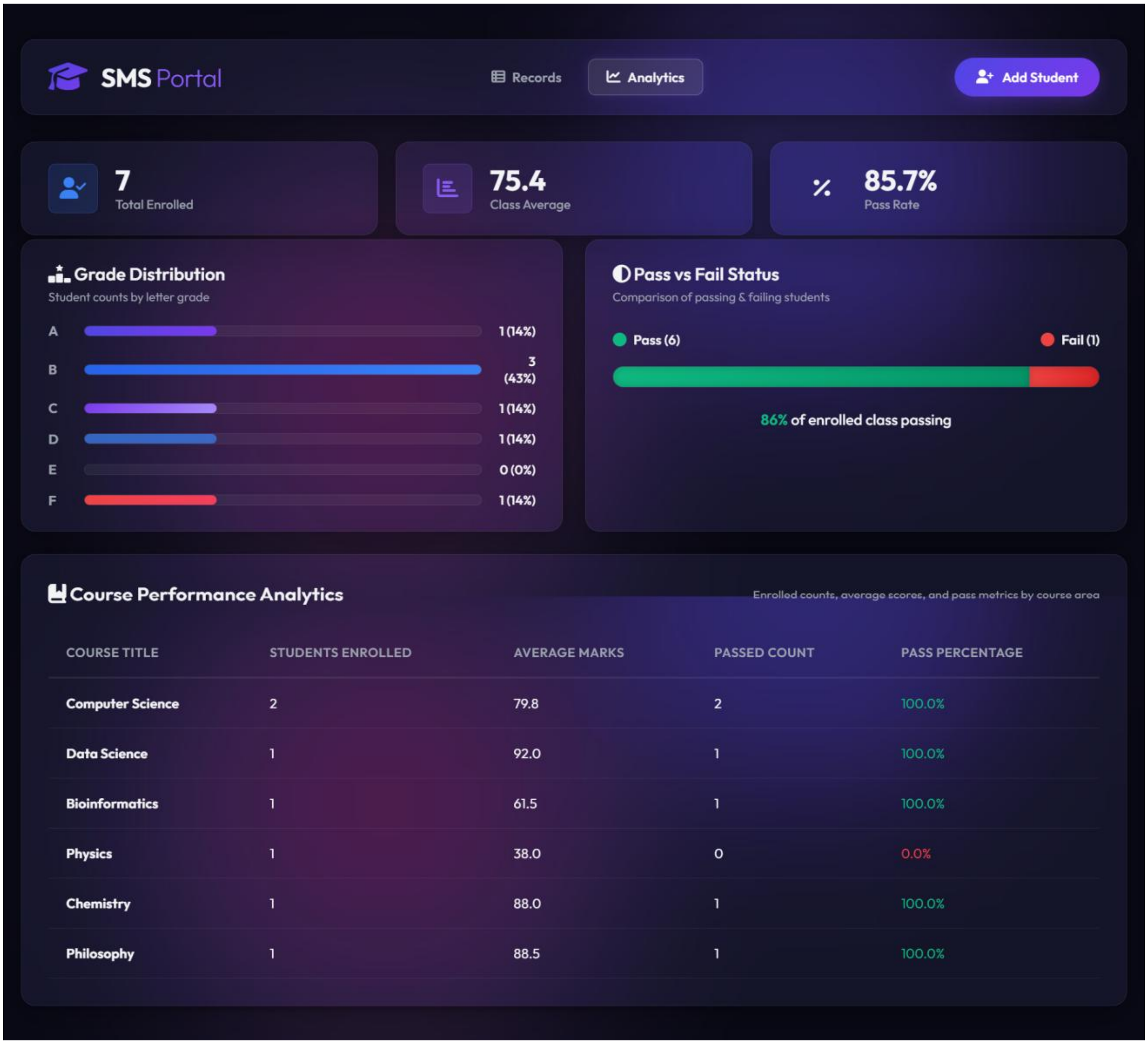


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces (if applicable)




SMS Portal
Records
Analytics
Add Student
7
Total Enrolled
75.4
Class Average
85.7%
Pass Rate
Grade Distribution
Student counts by letter grade
A 1 (14%)
B 3 (43%)
C 1 (14%)
D 1 (14%)
E 0 (0%)
F 1 (14%)
Pass vs Fail Status
Comparison of passing & failing students
Pass (6)
Fail (1)
86% of enrolled class passing
Course Performance Analytics
Enrolled counts, average scores, and pass metrics by course area
COURSE TITLE
STUDENTS ENROLLED
AVERAGE MARKS
PASSED COUNT
PASS PERCENTAGE
Computer Science 2 79.8 2 100.0%
Data Science 1 92.0 1 100.0%
Bioinformatics 1 61.5 1 100.0%
Physics 1 38.0 0 0.0%
Chemistry 1 88.0 1 100.0%
Philosophy 1 88.5 1 100.0%

6 Performance Test

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Result
Add Student	Valid data	Student Added
Add Duplicate ID	Existing ID	Error Message
Invalid Email	Wrong Email	Validation Error
Marks >100	110	Validation Error
Delete Student	Valid ID	Student Deleted
Search Student	Existing ID	Student Found

6.2 Test Procedure

1. Launch Flask Application.
2. Execute CRUD operations.
3. Run automated unit tests.
4. Validate API responses.
5. Verify JSON persistence.
6. Check frontend functionality.

6.3 Performance Outcome

The system successfully handled all CRUD operations with accurate data validation and persistence. Automated test cases passed successfully, confirming the correctness of validation rules and business logic.

The search operation demonstrated efficient lookup performance due to unique student ID indexing. Sorting operations correctly arranged records alphabetically and by marks. The web interface remained responsive during testing and successfully communicated with the Flask backend through REST API endpoints.

The project achieved its objective of providing a reliable, maintainable, and user-friendly student record management solution.

7 My learnings

During the development of the Student Management System, I gained valuable practical experience in various areas of software development. The project helped me strengthen my understanding of Object-Oriented Programming (OOP) concepts such as classes, objects, encapsulation, and modular design. I learned how to develop web applications using the Flask framework and gained hands-on experience in designing and implementing RESTful APIs for communication between the frontend and backend. Working with JSON-based file storage enhanced my knowledge of data persistence and file handling techniques. Additionally, I improved my frontend development skills by creating responsive user interfaces using HTML, CSS, and JavaScript. The project also introduced me to software testing using Python's unit test framework, enabling me to verify the correctness and reliability of the application. Throughout the development process, I enhanced my debugging, analytical thinking, and problem-solving abilities by identifying and resolving various technical challenges. Furthermore, I gained practical exposure to version control systems using Git and GitHub, which helped me manage source code effectively and collaborate using industry-standard development practices.

8 Future work scope

The Student Management System has been designed with a modular architecture, allowing it to be easily extended with additional features in the future. One of the major enhancements would be the integration of a relational database such as SQLite, MySQL, or PostgreSQL to improve scalability and data management. User authentication and role-based access control can be implemented to provide secure access for administrators, teachers, and students. Additional modules such as attendance management, fee management, examination and result processing, and course enrolment can further increase the system's functionality. The application can also be enhanced with advanced reporting features, including PDF and Excel export capabilities, along with interactive data visualization dashboards. Cloud deployment can make the system accessible from anywhere, while a dedicated mobile application can improve user convenience. Furthermore, modern technologies such as Artificial Intelligence and Machine Learning can be incorporated to analyze student performance, predict academic outcomes, and provide personalized recommendations. These enhancements would transform the current system into a comprehensive educational management platform suitable for real-world institutional use.